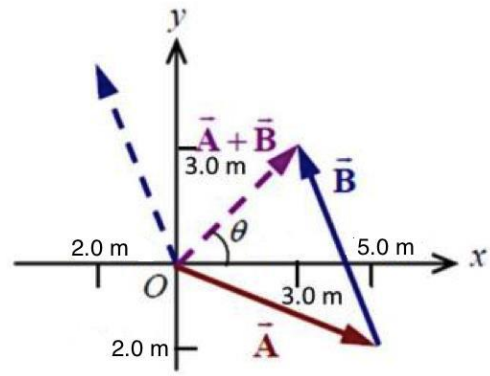
PHY101: Introduction to Physics-1

Monsoon, 2018

Tutorial-3



1. A group of vectors are shown in the adjoining figure. Find the direction and magnitude of (a) $\vec{A}$, (b) $\vec{B}$, and (c) $\vec{A}+\vec{B}$. For part (c) use both the triangle addition rule and component-resolution method and see if the results are consistent.

2. Two swimmers leave point *A* on one bank of the river to reach point *B* lying right across on the other bank. One of them crosses the river along the straight line *AB* while the other swims at right angles to the stream and then walks the distance that he has been carried away by the stream to get to point *B*. What was the velocity *u* of his walking if both swimmers reached the destination simultaneously? The stream velocity v_0 = 2.0 km/hour and the velocity v' of each swimmer with respect to water equals to 2.5 km per hour.

3. Find the work done in moving an object along a straight line from (3, 2, -1) to (2, -1, 4) in a force field given by **F** = 4**i** - 3**j** + 2**k**.

4. The velocity of a particle moving in two-dimensional space is given by,

$$\vec{v}(t) = -a\,\omega\sin(\omega\,t)\,\hat{\imath} + a\,\omega\cos(\omega\,t)\,\hat{\jmath}$$

where, a and ω are constants of appropriate units. At t = 0, the particle is at the position, $\vec{r}(t=0) = a\hat{\imath}$. Find out its position, $\vec{r}(t)$, at an arbitrary time instant t, expressed in Cartesian as well as plane-polar coordinates. How does the motion look like?

5. A point object starts rotating about a stationary axis with an angular acceleration of $\alpha = at$ where a = 0.02 rad/s^3. How soon after the beginning of rotation will the total acceleration vector of the object form an angle 60° with its velocity vector?